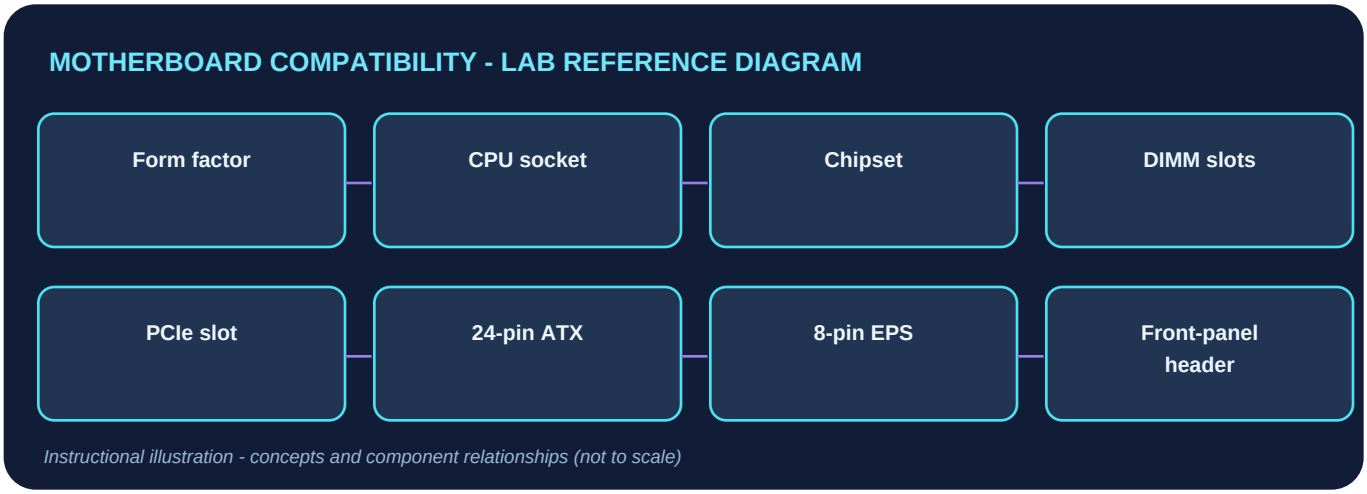
2.2.11 Choose and Install a Motherboard

STUDENT STUDY GUIDE + VOCABULARY + VISUAL REFERENCE

Learning goal

Select and install compatible PC hardware.

Reference image



Use the labeled illustration to identify the key components, commands, or checkpoints you will encounter in this lab.

Prepare for the simulation

1. Check case size, processor socket and RAM support.
2. Fit standoffs and install the I/O shield if required.
3. Secure the motherboard without overtightening.
4. Attach power, front-panel and storage connections.

Troubleshooting and safe practice

• No power: verify front-panel power switch pins.

• No POST: check CPU power and memory compatibility.

2.2.11 | VOCABULARY & REVIEW

Essential vocabulary

Term	Meaning / recognition cue
Form factor	A key term in motherboard compatibility; identify its purpose, where it is used, and how to verify it.
CPU socket	A key term in motherboard compatibility; identify its purpose, where it is used, and how to verify it.
Chipset	A key term in motherboard compatibility; identify its purpose, where it is used, and how to verify it.
DIMM slots	Motherboard connectors for desktop memory modules.
PCIe slot	Physical interface or connection point; verify its intended function and compatibility before use.
24-pin ATX	Primary motherboard power connector.
8-pin EPS	A key term in motherboard compatibility; identify its purpose, where it is used, and how to verify it.
Front-panel header	Physical interface or connection point; verify its intended function and compatibility before use.

Knowledge check - answer in your own words

1. Identify two components or commands in the reference diagram and explain what each does.

2. What should you verify before changing or installing hardware, settings, or files in this lab?

3. Choose one listed failure symptom. What is your first safe diagnostic step?

4. How will you prove the task is complete, rather than merely assuming success?

Lab readiness checklist

[ ] I can identify the labeled reference items. [ ] I understand the safety precautions.

[ ] I can explain the vocabulary. [ ] I can verify and document successful results.

Student: _____ Date: _____